

```

strResult1 = strResult1 & " " &
Word(10000000)
strX1 = Mid(strX1, 4, intLen)

Case 9:

strX2 = Mid(strX1, 1, 2)
strResult1 = strResult1 & " " &
TwoDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(10000000)
strX1 = Mid(strX1, 3, intLen)
.
.
Case 1:
strResult1 = strResult1 & " " &
Word(strX1)
strX1 = "0"

```

It works step-by-step as follows:

1. Round the given input to two decimal places
2. Extract the integer part of the rounded number, convert it into a string, i.e., 9178434685, and find its length, which is 10. (Refer the sequence of program execution listed in the code.)
3. Execute the 'do while' loop to check the length of the string (intLen>0):
- (i) Execute the block of 'case' state-

**TABLE III**  
**Grouping of the Digits for the Example**

| StrX1      | IntLen | Grouped digit | Position name |
|------------|--------|---------------|---------------|
| 9178434685 | 10     | 917           | Crore         |
| 8434685    | 7      | 84            | Lakh          |
| 34685      | 5      | 34            | Thousand      |
| 685        | 3      | 6             | Hundred       |

ment associated with the length of the string

(ii) Now, find the word equivalent of the required grouped digit from left side (in this case, the group is '917'), concatenate it with position name (see Table III) and append it to the output string. As a maximum of three digits are grouped, a separate function, ThreeDigitToWord, is required

(iii) The string is replaced with a new string by eliminating the first grouped digit from left side

(iv) Go to step (i)

4. Find the position of the decimal separator (.) of the original rounded number if existing

5. If position > 0, extract two digits from right of the decimal separator. Check whether its integer value is greater than 0. If yes, find its word

equivalent and append it to the output string  
6. Return the output

## Customisation

The program can be customised as per your requirement. For example, if Rs 9178434685.50 is to

be read as USD 9178434685.50, certain changes will be required in the code. The number for US currency will be read as 'dollars nine billion one seventy eight million four hundred thirty four thousand six hundred eighty five and cents fifty only.' Here grouping of the number and the way you read it out has changed, hence user-defined functions such as 'mid()' have to be changed in each 'case' statement in the above functions and 'paise' has to be replaced with 'cents'.

In the code, you can also replace the currency name with the item name before the output string, so you can use the same function for returning the item quantity in words.

**EFY note.** The source code of this program has been included in this month's EFY-CD.

## NUMWORD.FRM

```

Function ToWords(ByVal varNum As Variant) As String
Dim intLen As Integer, strX1 As String
Dim strX2 As String, strResult1 As String
Dim intPos As Integer

' round the number to two decimal places
varNum = Round(varNum, 2)

'convert integer part of number to string for easy processing
strX1 = CStr(Int(varNum))

'find the length of the string
intLen = Len(strX1)

'initialise the variable
strResult1 = ""

'do while length of string is > 0
Do While (intLen > 0)
intLen = Len(strX1)
Select Case intLen

```

```

Case 10:
strX2 = Mid(strX1, 1, 3)
strResult1 = strResult1 & " " &
ThreeDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(100000000)
strX1 = Mid(strX1, 4, intLen)

' comment above four lines and uncomment below four lines if currency is in dollars
strX2 = Mid(strX1, 1, 1)
strResult1 = strResult1 & " " &
Word(strX2)
strResult1 = strResult1 & " " &
Word(1000000000)
strX1 = Mid(strX1, 2, intLen)

Case 9:
strX2 = Mid(strX1, 1, 2)
strResult1 = strResult1 & " " &
TwoDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(100000000)
strX1 = Mid(strX1, 3, intLen)

' comment above four lines and uncomment below four lines if currency is in dollars
strX2 = Mid(strX1, 1, 3)
strResult1 = strResult1 & " " &

```

```

ThreeDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(10000000)
strX1 = Mid(strX1, 4, intLen)

Case 8:
strX2 = Mid(strX1, 1, 1)
strResult1 = strResult1 & " " &
Word(strX2)
strResult1 = strResult1 & " " &
Word(100000000)
strX1 = Mid(strX1, 2, intLen)

' comment above four lines and uncomment below four lines if currency is in dollars
strX2 = Mid(strX1, 1, 2)
strResult1 = strResult1 & " " &
TwoDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(10000000)
strX1 = Mid(strX1, 3, intLen)

Case 7:
strX2 = Mid(strX1, 1, 2)
strResult1 = strResult1 & " " &
TwoDigitToWord(strX2)
strResult1 = strResult1 & " " &
Word(1000000)
strX1 = Mid(strX1, 3, intLen)

```